Chapter 7

Expectation and Covariance

7.1 Expected value of functions of random variables X and Y

Suppose the random variable $Z = g(X, Y)$ is a function of two random variables X and Y and we are interested in computing $\mathbb{E}[Z]$. As with functions of a single random variable, there are two ways to compute $\mathbb{E}(Z)$. One way is to first find the probability mass function (discrete case) or probability density function (continuous case) of Z, and then compute $\mathbb{E}(Z)$ from this, as follows.

Discrete case: If X and Y are discrete random variables, then so is the random variable $Z = g(X, Y)$ and you can use the joint probability mass function $p_{X,Y}(x, y)$ to compute the probability mass function $p_Z(z)$ for Z. Once you have determined $p_Z(z)$, you can use this to compute $\mathbb{E}(Z)$:

$$\mathbb{E}(Z) = \sum_{Z \in \mathrm{Im} Z} z p_Z(z)$$

Continuous case: If X and Y are jointly continuous random variables with a joint density $f_{X,Y}(x, y)$ and the function $g : \mathbb{R}^2 \to \mathbb{R}$ is continuous, then the random variable Z is a continuous random variable, and you can use the joint density $f_{X,Y}(x, y)$ to determine $F_Z(z)$, the distribution function for Z. Then, to obtain the density $f_Z(z)$, differentiate $F_Z(z)$. Once you have determined $f_Z(z)$, you can use this to compute

$$\mathbb{E}(Z) = \int_{\mathbb{R}} z f_Z(z) \, \mathrm{d}z$$

The second way to compute $\mathbb{E}(Z)$ is more direct, without needing to first find the probability mass function or density for Z. If you do not need to compute probabilities for the random variable Z, then you do not need to determine the probability mass function or density for Z, and instead you should compute $\mathbb{E}(Z)$ as follows.

Discrete case: If X and Y are discrete random variables with joint probability mass function $p_{X,Y}(x, y)$ then

$$\boxed{\mathbb{E}(Z) = \mathbb{E}\big(g(X, Y)\big) = \sum_{\substack{x \in \mathrm{Im} X \\ y \in \mathrm{Im} Y}} g(x, y) p_{X,Y}(x, y)}$$

Continuous case: If X and Y are continuous random variables with joint density $f_{X,Y}(x,y)$ then

$$\boxed{\mathbb{E}(Z) = \mathbb{E}\big(g(X,Y)\big) = \iint_{\mathbb{R}^2} g(x,y) f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y}$$

Example 6.11 continued. Suppose that X and Y have joint density function

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x>0,\ y>0,\ x+y<1 \\ 0 & \text{otherwise} \end{cases}$$

Let $Z = XY$. Then

$$\begin{aligned}
\mathbb{E}(Z) = \mathbb{E}(XY) &= \iint xy f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\
&= \int_{y=0}^{1} \int_{x=0}^{1-y} 60x^2y^3\ \mathrm{d}x\ \mathrm{d}y \\
&= \int_{y=0}^{1} \left[20x^3y^3\right]_{x=0}^{1-y}\ \mathrm{d}y \\
&= \int_{y=0}^{1} 20(1-y)^3y^3\ \mathrm{d}y \\
&= \int_{y=0}^{1} 20\left(y^3 - 3y^4 + 3y^5 - y^6\right)\mathrm{d}y \\
&= \left[5y^4 - 12y^5 + 10y^6 - \frac{20y^7}{7}\right]_{y=0}^{1} = \frac{1}{7} \approx 0.143
\end{aligned}$$

Note: We can also compute $\mathbb{E}(X)$ and $\mathbb{E}(Y)$ *without* first computing the marginal densities $f_X(x)$ and $f_Y(y)$ for X and Y as follows:

$$\mathbb{E}(X) = \iint_{\mathbb{R}^2} x f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \text{ and } \mathbb{E}(Y) = \iint_{\mathbb{R}^2} y f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

Two Important Results

1. Suppose X and Y are *any* two random variables, and $a, b \in \mathbb{R}$. Then

$$\boxed{\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)}$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *any* random variables, and $a_1, a_2, \ldots, a_n \in \mathbb{R}$, then

$$\mathbb{E}(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1\mathbb{E}(X_1) + a_2\mathbb{E}(X_2) + \cdots + a_n\mathbb{E}(X_n)$$

2. Suppose X and Y are *independent* random variables. Then

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *independent* random variables, then

$$\mathbb{E}(X_1X_2 \ldots X_n) = \mathbb{E}(X_1)\mathbb{E}(X_2) \ldots \mathbb{E}(X_n)$$

Proof. We will prove these two results in the case where X and Y are jointly continuous with density $f_{X,Y}(x,y)$, but the results are true regardless of whether X, Y are discrete or continuous or not.

1. With X, Y jointly continuous, we have

$$\begin{aligned}\mathbb{E}(aX+bY) &= \iint_{\mathbb{R}^2}(ax+by)f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\ &= \iint_{\mathbb{R}^2} axf_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y + \iint_{\mathbb{R}^2} byf_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y \\ &= \int_{\mathbb{R}} ax\left(\int_{\mathbb{R}} f_{X,Y}(x,y)\ dy\right)dx + \int_{\mathbb{R}} by\left(\int_{\mathbb{R}} f_{X,Y}(x,y)\ dx\right)dy \\ &= a\int_{\mathbb{R}} xf_X(x)\ \mathrm{d}x + b\int_{\mathbb{R}} yf_Y(y)\ \mathrm{d}y \\ &= a\mathbb{E}(X)+b\mathbb{E}(Y)\end{aligned}$$

2. With X, Y jointly continuous and *independent*, we have

$$\begin{aligned}\mathbb{E}(XY) &= \iint_{\mathbb{R}^2} xyf_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{y=-\infty}^{\infty}\int_{x=-\infty}^{\infty} xyf_X(x)f_Y(y)\,\mathrm{d}x\,\mathrm{d}y \qquad \text{(since } X,\ Y \text{ are independent)} \\ &= \int_{y=-\infty}^{\infty} yf_Y(y)\left(\int_{x=-\infty}^{\infty} xf_X(x)\mathrm{d}x\right)\mathrm{d}y \\ &= \int_{y=-\infty}^{\infty} yf_Y(y)\cdot\mathbb{E}(X)\,\mathrm{d}y \\ &= \mathbb{E}(X)\left(\int_{y=-\infty}^{\infty} yf_Y(y)\,\mathrm{d}y\right) = \mathbb{E}(X)\mathbb{E}(Y)\end{aligned}$$

Remarks

1. The calculations which establish these result in the case where X and Y are *discrete* random variables are similar. You can do them as an exercise.

2. If $X_1, X_2, \ldots, X_n$ are any random variables, and $g_1, g_2, \ldots, g_n$ are functions, then

$$\mathbb{E}\big(g_1(X_1)+g_2(X_2)+\cdots+g_n(X_n)\big)=\mathbb{E}\big(g_1(X_1)\big)+\mathbb{E}\big(g_2(X_2)\big)+\cdots+\mathbb{E}\big(g_n(X_n)\big)$$

3. If $X_1, X_2, \ldots, X_n$ are (jointly) *independent* random variables, then so are the random variables $g_1(X_1), g_2(X_2), \ldots, g_n(X_n)$ where for $k = 1, 2, ..., n$, $g_k : \mathbb{R} \to \mathbb{R}$ is function.
So we also have

$$\mathbb{E}\big(g_1(X_1)g_2(X_2)\ldots g_n(X_n)\big)=\mathbb{E}\big(g_1(X_1)\big)\mathbb{E}\big(g_2(X_2)\big)\ldots\mathbb{E}\big(g_n(X_n)\big)$$

7.2 Covariance and correlation

Definition
Suppose X is a random variable with expected value $\mathbb{E}(X) = \mu_X$ and Y is a random variable with expected value $\mathbb{E}(Y) = \mu_Y$. Then $\mathbb{C}\mathrm{ov}(X, Y)$, the **covariance** of X and Y, is defined as

$$\boxed{\mathbb{C}\mathrm{ov}(X,Y)=\mathbb{E}\big((X-\mu_X)(Y-\mu_Y)\big)}$$

Remark One reason the covariance is useful is that its *sign* tells us something about how the variables X and Y are associated:

- If $\mathbb{C}\text{ov}(X,Y) > 0$ then we say that X and Y are *positively associated*. If the random variables X and Y are positively associated, then *larger* values of X tend to be associated with *larger* values of Y.
  Example: Suppose that X = an individual's height, Y = the same individual's weight, then we would expect $\mathbb{C}\text{ov}(X,Y) > 0$.

- If $\mathbb{C}\text{ov}(X,Y) < 0$ then we say that X and Y are *negatively associated*. If the random variables X and Y are negatively associated, then *larger* values of X tend to be associated with *smaller* values of Y (and vice versa).
  Example: Suppose that X = heating fuel consumption in a house and Y = outside temperature, then we would expect $\mathbb{C}\text{ov}(X,Y) < 0$.

If $\mathbb{C}\text{ov}(X,Y) = 0$, we say that X and Y are **uncorrelated**.

Important simplification:
Using the properties of expectation, we obtain

$$\begin{aligned}\mathbb{C}\text{ov}(X,Y) &= \mathbb{E}\big((X-\mu_X)(Y-\mu_Y)\big) = \mathbb{E}(XY - \mu_X Y - \mu_Y X + \mu_X\mu_Y) \qquad \text{(algebra)}\\ &= \mathbb{E}(XY) - \mu_X\mathbb{E}(Y) - \mu_Y\mathbb{E}(X) + \mu_X\mu_Y \qquad \text{(by properties of expectation)}\\ &= \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)\end{aligned}$$

Remark: We usually *compute* $\mathbb{C}\text{ov}(X,Y)$ using $\mathbb{C}\text{ov}(XY) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$.

Special case
If random variables X and Y are *independent*, then

$$\mathbb{C}\text{ov}(X,Y) \overset{\text{(from above)}}{=} \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) \overset{\text{(independence)}}{=} \mathbb{E}(X)\mathbb{E}(Y) - \mathbb{E}(X)\mathbb{E}(Y) = 0,$$

i.e. if the random variables X and Y are *independent*, then X and Y are *uncorrelated*.
Watch out: The *converse* of this is **not true**. That is, $\mathbb{C}\text{ov}(X,Y) = 0$ *does not imply* that the variables X and Y are independent.

Properties of $\mathbb{C}\text{ov}(X,Y)$: It is straightforward (*exercise*) to show that

1. $\mathbb{C}\text{ov}(X,Y) = \mathbb{C}\text{ov}(Y,X)$

2. $\mathbb{C}\text{ov}(X+Y,Z) = \mathbb{C}\text{ov}(X,Z) + \mathbb{C}\text{ov}(Y,Z)$

3. $\mathbb{C}\text{ov}(aX+b,cY+d) = ac\mathbb{C}\text{ov}(X,Y)$

Discussion
The *sign* of $\mathbb{C}\text{ov}(X,Y)$ gives an indication of the 'direction' of the association between X and Y (i.e. whether there is positive or negative association), but the numerical value of $\mathbb{C}\text{ov}(X,Y)$ is *not* a good measure of the *strength* of the association between the random variables X and Y. The problem is that $\mathbb{C}\text{ov}(X,Y)$ is not a *scale-free* measure of the association between X and Y. Here is an example that illustrates this problem.

Example 7.1. Suppose a student is selected at random and the student's height and foot length are measured. Based on these measurements, let

$X =$ student's height in metres
$Y =$ student's foot length in metres
$U =$ student's height in centimetres (so $U = 100X$)
$V =$ student's foot length in centimetres (so $V = 100Y$)

Then we have

$$\mathbb{C}\text{ov}(U, V) = \mathbb{C}\text{ov}(100X, 100Y) = 10000\mathbb{C}\text{ov}(X, Y)$$

So $\mathbb{C}\text{ov}(U, V)$ is ten thousand times greater than $\mathbb{C}\text{ov}(X, Y)$, but the 'strength' of the association between 'height' and 'foot length' should not depend on the *scale* that we use to measure these physical quantities.

Conclusion:
The example above illustrates why we need a measure of the association between variables that does not depend on the *scale* that we use for the variables. A *scale-free* measure of the strength of the association between two random variables X and Y is provided by the *correlation coefficient*.

Definition 7.1. The *correlation coefficient* $\rho(X, Y)$ for X and Y is defined by

$$\boxed{\rho(X, Y) = \frac{\mathbb{C}\text{ov}(X, Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}} = \frac{\mathbb{C}\text{ov}(X, Y)}{\sigma_X \sigma_Y}}$$

Here σ_X is the standard deviation of X and σ_Y is the standard deviation of Y.
Example 7.1 continued. For the height and foot length variables in example 7.1,

$$\begin{aligned}
\rho(U, V) &= \frac{\mathbb{C}\text{ov}(U, V)}{\sqrt{\mathbb{V}\text{ar}(U)\mathbb{V}\text{ar}(V)}} = \frac{\mathbb{C}\text{ov}(100X, 100Y)}{\sqrt{\mathbb{V}\text{ar}(100X)\mathbb{V}\text{ar}(100Y)}} \\
&= \frac{(100)^2\mathbb{C}\text{ov}(X, Y)}{\sqrt{(100)^2\mathbb{V}\text{ar}(X)(100)^2\mathbb{V}\text{ar}(Y)}} \qquad \text{(by the properties of variance and covariance)} \\
&= \frac{\mathbb{C}\text{ov}(X, Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}} = \rho(X, Y)
\end{aligned}$$

So: $\rho(U, V) = \rho(X, Y)$. In other words, the *correlation* between 'height' and 'foot length' does not depend on the *scale* used to measure these quantities.

Properties of the correlation coefficient $\rho(X, Y)$:

1. $\rho(aX + b, cY + d) = \rho(X, Y) \quad \forall a, b, c, d \in \mathbb{R}$ s.t. $ac > 0$.
   (i.e $\rho(X, Y)$ is scale and location invariant)

2. Whenever $\rho(X, Y)$ exists (i.e. $\mathbb{V}\text{ar}(X), \mathbb{V}\text{ar}(Y) \neq 0$) then

   $$-1 \leqslant \rho(X, Y) \leqslant 1.$$

   Also if $\rho(X, Y) = \pm 1$, then $Y = aX + b$ for some $a, b \in \mathbb{R}$.

Proof.

1.
$$\begin{aligned}\rho(aX+b,cY+d) &= \frac{\mathbb{C}\text{ov}(aX+b,cY+d)}{\sqrt{\mathbb{V}\text{ar}(aX+b)\mathbb{V}\text{ar}(cY+d)}}\\ &= \frac{ac\mathbb{C}\text{ov}(X,Y)}{\sqrt{a^2\mathbb{V}\text{ar}(X)c^2\mathbb{V}\text{ar}(Y)}}\\ &= \frac{\mathbb{C}\text{ov}(X,Y)}{\sqrt{\mathbb{V}\text{ar}(X)\mathbb{V}\text{ar}(Y)}}\\ &= \rho(X,Y)\end{aligned}$$

2. Let $\hat{X} = \frac{X-\mathbb{E}(X)}{\sqrt{\mathbb{V}\text{ar}(X)}}$, $\hat{Y} = \frac{Y-\mathbb{E}(Y)}{\sqrt{\mathbb{V}\text{ar}(Y)}}$. Then it follows from Property 1 above that
$$\rho(X,Y) = \rho(\hat{X},\hat{Y}) = \mathbb{E}(\hat{X}\hat{Y}) \qquad (\text{since } \mathbb{E}(\hat{X}) = \mathbb{E}(\hat{Y}) = 0)$$
Next we note that
$$0 \leqslant \mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = \mathbb{E}(\hat{X}^2) - 2\mathbb{E}(\hat{X}\hat{Y}) + \mathbb{E}(\hat{Y}^2) = 2 - 2\mathbb{E}(\hat{X}\hat{Y}) \qquad (*)$$
since $\mathbb{E}(\hat{X}^2) = \mathbb{E}(\hat{Y}^2) = 1$. So, it follows from $(*)$ that $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) \leqslant 1$.

Similarly, we have
$$0 \leqslant \mathbb{E}\Big(\big(\hat{X}+\hat{Y}\big)^2\Big) = \mathbb{E}(\hat{X}^2) + 2\mathbb{E}(\hat{X}\hat{Y}) + \mathbb{E}(\hat{Y}^2) = 2 + 2\mathbb{E}(\hat{X}\hat{Y})$$
So we must have $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) \geqslant -1$. It follows that $-1 \leq \rho(X,Y) \leq 1$.

Next, suppose $\rho(X,Y) = \mathbb{E}(\hat{X}\hat{Y}) = 1$. Then it follows from $(*)$ that $\mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = 0$. We note that since $\mathbb{E}\Big(\big(\hat{X}-\hat{Y}\big)^2\Big) = 0$, we must also have $\hat{X} = \hat{Y}$. In other words, by definition of $\hat{X}$ and $\hat{Y}$, we have
$$\frac{X-\mathbb{E}(X)}{\sqrt{\mathbb{V}\text{ar}(X)}} = \frac{Y-\mathbb{E}(Y)}{\sqrt{\mathbb{V}\text{ar}(Y)}} \qquad (**).$$
Solving the equation $(**)$ for Y, we obtain
$$Y = \underbrace{\frac{\sqrt{\mathbb{V}\text{ar}(Y)}}{\sqrt{\mathbb{V}\text{ar}(X)}}}_{a} X + \underbrace{\mathbb{E}(Y) - \mathbb{E}(X)\frac{\sqrt{\mathbb{V}\text{ar}(Y)}}{\sqrt{\mathbb{V}\text{ar}(X)}}}_{b}$$
as desired.
Exercise: Show that if $\rho(X,Y) = -1$, then $Y = aX + b$ for some $a, b \in \mathbb{R}$.

7.3 Variance of a linear sum of random variables X and Y

In this section we investigate the variance of a linear sum of random variables. We already know that for random variables X and Y and $a, b \in \mathbb{R}$
$$\mathbb{E}(aX+bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

Using the properties of variance and the definition of the covariance, we can now determine the corresponding result for $\mathbb{V}\text{ar}(aX + bY)$:

$$\begin{aligned}\mathbb{V}\text{ar}(aX + bY) &= \mathbb{E}\big((aX + bY)^2\big) - \big(\mathbb{E}(aX + bY)\big)^2 \\ &= a^2\mathbb{E}(X^2) + 2ab\mathbb{E}(XY) + b^2\mathbb{E}(Y^2) - a^2\big(\mathbb{E}(X)\big)^2 - 2ab\mathbb{E}(X)\mathbb{E}(Y) - b^2\big(\mathbb{E}(Y)\big)^2 \\ &= a^2\mathbb{V}\text{ar}(X) + 2ab\mathbb{C}\text{ov}(X, Y) + b^2\mathbb{V}\text{ar}(Y) \qquad (**)\end{aligned}$$

In particular,

$$\boxed{\mathbb{V}\text{ar}(X + Y) = \mathbb{V}\text{ar}(X) + 2\mathbb{C}\text{ov}(X, Y) + \mathbb{V}\text{ar}(Y)}$$

Remarks:

1. The above result shows that to compute $\mathbb{V}\text{ar}(X+Y)$ it is *not* enough to know the variances of each of the variables X and Y. We also need to know the covariance $\mathbb{C}\text{ov}(X, Y)$.

2. If X and Y are *independent* then $\mathbb{C}\text{ov}(X, Y) = 0$ and equation $(**)$ above becomes

   $$\mathbb{V}\text{ar}(aX + bY) = a^2\mathbb{V}\text{ar}(X) + b^2\mathbb{V}\text{ar}(Y).$$

   More generally, if $X_1, X_2, \ldots, X_n$ are *independent*, then

   $$\mathbb{V}\text{ar}(a_1X_1 + a_2X_2 + \cdots + a_nX_n) = a_1^2\mathbb{V}\text{ar}(X_1) + a_2^2\mathbb{V}\text{ar}(X_2) + \cdots + a_n^2\mathbb{V}\text{ar}(X_n)$$

   and, in particular,

   $$\mathbb{V}\text{ar}(X_1 + \cdots + X_n) = \mathbb{V}\text{ar}(X_1) + \cdots + \mathbb{V}\text{ar}(X_n) \text{ for } \textit{independent } X_1, X_2, \ldots, X_n$$

Example 7.2. Suppose X is the number of successes in n independent, identical trials with $\mathbb{P}(\text{'success'}) = p$ for each trial. That is, X is a binomial random variable, $X \sim \text{Bin}(n, p)$, with probability mass function $p_X(k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $k = 0, 1, 2, ..., n$.
In chapter 3, we used the probability mass function to compute $\mathbb{E}(X)$; we can now compute $\mathbb{E}(X)$ much more easily using properties of expectation.

Recall that if we define the indicator random variables $I_1, I_2, \ldots, I_n$ to be

$$I_k = \begin{cases} 1 & \text{if } k^{\text{th}} \text{ trial is success (with probability } p) \\ 0 & \text{otherwise (with probability } 1-p) \end{cases}$$

then we can write X as the sum of these indicator random variables:

$$X = I_1 + I_2 + \cdots + I_n$$

and for $k = 1, 2, \ldots, n$, we have

$$\begin{aligned} \mathbb{E}(I_k) &= p \times 1 + (1-p) \times 0 = p \\ \mathbb{E}(I_k^2) &= p \times 1^2 + (1-p) \times 0^2 = p \\ \Rightarrow \quad \mathbb{V}\text{ar}(I_k) &= p - p^2 = p(1-p) \end{aligned}$$

So

$$\begin{aligned} \mathbb{E}(X) &= \mathbb{E}(I_1 + I_2 + \cdots + I_n) \\ &= \mathbb{E}(I_1) + \mathbb{E}(I_2) + \cdots + \mathbb{E}(I_n) \\ &= p + p + \cdots + p \\ &= np \end{aligned}$$

Note that we have not yet made use of independence. We will now assume that the indicator variables $I_1, I_2, \ldots, I_n$ are *independent* because the trials that the indicators are based on are physically independent trials. Then

$$\begin{aligned}
\mathbb{V}\text{ar}(X) &= \mathbb{V}\text{ar}(I_1 + I_2 + \cdots + I_n) \\
&= \mathbb{V}\text{ar}(I_1) + \mathbb{V}\text{ar}(I_2) + \cdots + \mathbb{V}\text{ar}(I_n) \qquad \text{(by independence of } I_1, I_2, \ldots, I_n\text{)} \\
&= p(1-p) + p(1-p) + \cdots + p(1-p) \\
&= np(1-p)
\end{aligned}$$

So $X \sim \text{Bin}(n,p) \Rightarrow \mathbb{E}(X) = np,\ \mathbb{V}\text{ar}(X) = np(1-p)$

Example 7.3. Statistical inference result. Suppose $X_1, X_2, \ldots X_n$ are i.i.d. (independent, identically distributed) random variables with

$$\mathbb{E}(X_k) = \mu \text{ and } \mathbb{V}\text{ar}(X_k) = \sigma^2 \quad \text{ for } k = 1, 2, ..., n.$$

In statistical inference we use the sample mean

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

to *estimate* the (population) mean μ.

The sample mean $\bar{X}$ is *also* a random variable, and it is important to know $\mathbb{E}(\bar{X})$ and $\mathbb{V}\text{ar}(\bar{X})$, the expected value and variance of the sample mean.

$$\begin{aligned}
\mathbb{E}(\bar{X}) &= \mathbb{E}\Big(\frac{X_1 + \cdots + X_n}{n}\Big) \\
&= \frac{\mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n)}{n} \qquad \text{(by linearity of expectation)} \\
&= \frac{\mu + \cdots + \mu}{n} = \frac{n\mu}{n} = \mu
\end{aligned}$$

Remark: This means that $\bar{X}$ is an *unbiased* estimator of μ.

$$\begin{aligned}
\mathbb{V}\text{ar}(\bar{X}) &= \mathbb{V}\text{ar}\Big(\frac{X_1 + \cdots + X_n}{n}\Big) \\
&= \frac{1}{n^2}\mathbb{V}\text{ar}(X_1 + \cdots + X_n) \qquad \text{(property of variance)} \\
&= \frac{1}{n^2}\big(\mathbb{V}\text{ar}(X_1) + \cdots + \mathbb{V}\text{ar}(X_n)\big) \qquad \text{(by independence of } X_1, X_2, \ldots X_n\text{)} \\
&= \frac{1}{n^2}(\sigma^2 + \cdots + \sigma^2) = \frac{\sigma^2}{n}
\end{aligned}$$

Remark: Notice that as $n \to \infty$, $\mathbb{V}\text{ar}(\bar{X}) \to 0$. In terms of statistical inference, this means that the larger the value of n (i.e. the more observations we have), the greater the precision of the estimator $\bar{X}$ for μ.